

Package ‘wbCorr’

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Title Bivariate Within- and Between-Cluster Correlations

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Description The wbCorr package separates the variance of variables provided into their within- and between-cluster components, and calculates the respective bivariate correlations for both levels separately. This is especially useful for longitudinal analyses. By centering the variables within and between clusters, this package helps users gain an initial understanding of their data before employing more sophisticated multilevel analyses. This approach allows for a more in-depth examination of relationships at both the within- and between-cluster levels, which can be essential for making informed decisions based on the structure of the data.

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Contents

get_ICC	2
get_matrix	3
get_table	4
plot,wbCorr-method	5
print,wbCorr-method	6
show,wbCorr-method	7
simdat_intensive_longitudinal	7
to_excel	8
update_wbCorr	9
wbCorr	9
wbCorr-class	12

Index	14
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get_ICC	<i>Return all ICCs for the original variables.</i>
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Description

You can use `get_ICC()` or `get_ICCs()` interchangeably.

Usage

```
get_ICC(object)
```

```
get_ICCs(object)
```

```
get_icc(object)
```

Arguments

`object` A `wbCorr` object, created by the `wbCorr()` function.

Value

A dataframe with ICCs for all variables. ICC is obtained by fitting mixed effects models and extracting the variance components. Then, the formula between- variance / total- variance is applied.

See Also

[wbCorr](#)

Examples

```
# importing our simulated example dataset with pre-specified within- and between- correlations
data("simdat_intensive_longitudinal")

# create object:
correlations <- wbCorr(data = simdat_intensive_longitudinal,
                       cluster = 'participantID')

# returns the ICCs:
```

```
ICCs <- get_ICC(correlations)
print(ICCs)
```

get_matrix	<i>Return matrices for within- and/or between-cluster correlations.</i>
------------	---

Description

You can use `summary()`, `get_matrices()`, or `get_matrix()` interchangeably. Merged matrices include the ICC on the diagonal. For more detailed statistics, use `get_table()`.

Usage

```
get_matrix(object, which = c("within", "between", "merge"), ...)

get_matrices(object, which = c("within", "between", "merge"), ...)

## S4 method for signature 'wbCorr'
summary(object, which = c("within", "between", "merge"), ...)
```

Arguments

object	A <code>wbCorr</code> object, created by the <code>wbCorr()</code> function.
which	A string or a character vector indicating which summaries to return. Options are 'within' or 'w', 'between' or 'b', and various merge options like 'merge', 'm', 'merge_wb', 'wb', 'merge_bw', 'bw'. Default is <code>c('within', 'between', 'merge')</code> .
...	Additional arguments passed to the base summary method

Value

A list containing the selected matrices of within- and/or between-cluster correlations, and ICCs on the diagonals for merged matrices.

See Also

[get_tables](#), [wbCorr](#)

Examples

```
# importing our simulated example dataset with pre-specified within- and between- correlations
data("simdat_intensive_longitudinal")

# create object:
correlations <- wbCorr(data = simdat_intensive_longitudinal,
                      cluster = 'participantID')

# returns a correlation matrix with stars for p-values:
matrices <- summary(correlations) # the get_matrix() and get_matrices() functions are equivalent
print(matrices)
```

```
# Access specific matrices by:
# Option 1:
matrices$within
# Option 2:
within_matrix <- summary(correlations, which = 'w') # or use 'within'
merged_within_between <- summary(correlations, which = 'wb')
print(within_matrix) # could be saved to an excel or csv file (e.g., write.csv)
```

get_table

Retrieve full tables for both within- and/or between-cluster correlations for a wbCorr object.

Description

This function has an alias `get_tables()` which can be used interchangeably. For correlations matrices, see the `summary()` function.

Usage

```
get_table(object, which = c("within", "between"))

get_tables(object, which = c("within", "between"))
```

Arguments

<code>object</code>	A <code>wbCorr</code> object, created by the <code>wbCorr()</code> function.
<code>which</code>	A character vector indicating which correlation table to return. Options are 'within' or 'w', and 'between' or 'b'.

Value

A list containing the selected tables of within- and/or between-cluster correlations.

See Also

[summary](#), [wbCorr](#)

Examples

```
# importing our simulated example dataset with pre-specified within- and between- correlations
data("simdat_intensive_longitudinal")

# create object:
correlations <- wbCorr(data = simdat_intensive_longitudinal,
                      cluster = 'participantID')

# returns a list with full detailed tables of the correlations:
tables <- get_table(correlations) # the get_tables() function is equivalent
print(tables)

# Access specific tables by:
# Option 1:
```

```

tables$between
# Option 2:
within_table <- get_tables(correlations, which = 'w') # or use 'within' or 'between'
print(within_table) # within_table could be saved to an excel or csv file (e.g., write.csv)

```

plot,wbCorr-method *Plot within- and between associations*

Description

Plots the centered variables of the provided dataframe against each other. Choose whether to plot the between-centered variables (representing the between-cluster correlations by plotting cluster means) or the within-centered variables (representing the within-cluster correlations by plotting deviations from person-means). A regression line is provided and the corresponding coefficient with significance displayed.

Usage

```

## S4 method for signature 'wbCorr'
plot(
  x,
  y,
  which = NULL,
  plot_NA = TRUE,
  standardize = TRUE,
  outlier_detection = "zscore",
  outlier_threshold = "recommended",
  type = "p",
  pch = 20,
  dot_lwd = 2,
  reg_lwd = 2,
  ...
)

```

Arguments

x	A wbCorr object to be plotted.
y	Choose which correlations to plot ('within' / 'w' or 'between' / 'b'); can be used as a positional argument.
which	Can be used as an alternative to 'y' (e.g., which = 'w'). It has the same functionality as 'y', but takes precedence if both are specified.
plot_NA	Boolean. Whether variables that have no variation on the selected level should be plotted or not.
standardize	Boolean. Whether the dataset should be standardized. If TRUE, the regression coefficient is equivalent to the pearson correlation.
outlier_detection	If FALSE, outliers will not be marked in red. Otherwise you may provide the method. Choose from: 'zscore', 'mad', or 'tukey'.

outlier_threshold	If 'recommended', the threshold for 'zscore' and 'mad' will be set to 3, and for 'tukey' to 1.5. You can provide and other numeric here.
type	points, lines, etc. see ?base::plot for available types).
pch	Graphical parameter. Select which type of points should be plotted.
dot_lwd	Graphical parameter. Set size of the points.
reg_lwd	Graphical parameter. Set thickness of the regression line.
...	further options to be passed to the base plot (pairs) function.

See Also[wbCorr](#)

print,wbCorr-method	<i>Print Method for the wbCorr Class</i>
---------------------	--

Description

Prints a summary of the wbCorr object.

Usage

```
## S4 method for signature 'wbCorr'
print(x, ...)
```

Arguments

x	A wbCorr object.
...	Additional arguments, currently unused.

See Also[wbCorr](#)**Examples**

```
# Example
data("simdat_intensive_longitudinal")
correlations <- wbCorr(simdat_intensive_longitudinal,
  cluster = 'participantID',
  confidence_level = 0.95,
  method = 'spearman',
  weighted_between_statistics = FALSE)

print(correlations)
```

show,wbCorr-method	<i>Show Method for the wbCorr Class</i>
--------------------	---

Description

Shows a summary of the wbCorr object, equivalent to the print method.

Usage

```
## S4 method for signature 'wbCorr'
show(object)
```

Arguments

object A wbCorr object.

See Also

[wbCorr](#), [print.wbCorr](#)

Examples

```
# Example using the iris dataset
cors <- wbCorr(iris, iris$Species, weighted_between_statistics = TRUE)
show(cors)
```

simdat_intensive_longitudinal	<i>Simulated Intensive Longitudinal Dataset</i>
-------------------------------	---

Description

A simulated intensive longitudinal dataset to test the package capabilities. This dataset contains 80 participants, day, and three variables (var1, var2, and var3) that are all correlated on both within- and between-levels.

Format

A data frame with the following columns:

participantID Identifier for each participant (integer)

day Day variable varying only within-person (integer)

var1 Variable 1 (numerical)

var2 Variable 2 (numerical)

var3 Variable 3 (numerical)

Details

The within-person correlations are all positive:

- var1 & var2: 0.1
- var1 & var3: 0.3
- var2 & var3: 0.8

The between-person correlations are all negative:

- var1 & var2: -0.5
- var1 & var3: -0.4
- var2 & var3: -0.2

Time trends (within):

- var1 & time: 0.0
- var2 & time: 0.0
- var3 & time: 0.4

Source

A simulated dataset by P. Küng

to_excel	<i>Saves the passed summary or table to excel</i>
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Description

Use `to_excel(get_matrix(wbCorrObject))` or `to_excel(get_table(wbCorrObject))` to save the provided table/matrix to an excel file.

Usage

```
to_excel(SummaryObject, path = file.path(getwd(), "wbCorr.xlsx"))
```

Arguments

SummaryObject	A summary or matrix object, such as those returned by <code>get_matrix()</code> or <code>get_table()</code> .
path	Specify the filename and a path. If no path is provided, the file will be saved to the current working directory.

Value

Writes an Excel file (.xlsx) to disk.

See Also

[get_tables](#), [wbCorr](#), [get_matrix](#)

Examples

```
# Importing our simulated example dataset with pre-specified within- and between- correlations
data("simdat_intensive_longitudinal")

# Create object:
correlations <- wbCorr(data = simdat_intensive_longitudinal,
                       cluster = 'participantID')

# Returns a correlation matrix with stars for p-values:
matrices <- get_matrix(correlations) # summary(correlations) works too.

to_excel(matrices)
```

update_wbCorr	<i>Check for updates of wbCorr</i>
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Description

This function checks if there is a newer version of a given R package on GitHub by comparing the version numbers in the local and remote DESCRIPTION files. It prints the current package version, the new version available on GitHub, and a message indicating whether an update is available.

Usage

```
update_wbCorr(ask = FALSE)
```

Arguments

ask	Logical, if set to TRUE, the function checks for updates and prompts the user to confirm the update when a newer version is found. If set to FALSE, the function automatically attempts to install the latest version without prompting when an update is available.
-----	--

Value

An integer: 1 if there's a newer version available, 0 if the current version is the latest, or NULL if there was an error accessing the remote DESCRIPTION file.

wbCorr	<i>wbCorr</i>
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Description

The wbCorr function creates a wbCorr object containing within- and between-cluster correlations, p-values, and confidence intervals for a given dataset and clustering variable. The object can be plotted.

Usage

```

wbCorr(
  data,
  cluster,
  confidence_level = 0.95,
  method = "pearson",
  bootstrap = FALSE,
  nboot = 1000,
  weighted_between_statistics = NULL,
  between_weighting = c("equal_clusters", "cluster_size"),
  between_inference = c("analytic", "none"),
  centering_rows = c("pairwise_complete", "all_available")
)

wbcorr(
  data,
  cluster,
  confidence_level = 0.95,
  method = "pearson",
  bootstrap = FALSE,
  nboot = 1000,
  weighted_between_statistics = NULL,
  between_weighting = c("equal_clusters", "cluster_size"),
  between_inference = c("analytic", "none"),
  centering_rows = c("pairwise_complete", "all_available")
)

```

Arguments

data	A dataframe containing numeric variables for which correlations will be calculated.
cluster	A vector representing the clustering variable or a string with the name of the column in data that contains the clustering variable.
confidence_level	A numeric value between 0 and 1 representing the desired level of confidence for confidence intervals (default: 0.95).
method	A string indicating the correlation method to be used. Supported methods are 'pearson', 'spearman', and 'spearman-jackknife'. (default: 'pearson'). 'pearson': Pearson correlation method uses t-statistics to determine confidence intervals and p-values. 'spearman': Spearman correlation method uses the Fisher z-transformation for confidence intervals and p-values. 'spearman-jackknife': Employs the Euclidean jackknife technique to compute confidence intervals, providing more robust confidence intervals in the presence of non-normal data or outliers. Note that p-values are not available when this method is selected.
bootstrap	Performs a bias-corrected and accelerated (BCa) parametric bootstrap to compute confidence intervals and p-values. Recommended for non-normal data, but slow. (default: FALSE).
nboot	Specifies the amount of bootstrap samples (default: 1000).
weighted_between_statistics	Deprecated logical alias for between_weighting. If TRUE, between_weighting = "cluster_size"; if FALSE, between_weighting = "equal_clusters".

between_weighting

A string specifying the between-cluster estimand. "equal_clusters" correlates pair-specific cluster means with each cluster contributing equally. "cluster_size" computes a sample-size weighted correlation of pair-specific cluster means, using the number of complete observation pairs in each cluster as weights.

between_inference

A string specifying whether between-cluster p-values and confidence intervals are calculated analytically ("analytic") or omitted ("none"). Analytic inference for "cluster_size" weighted between correlations uses $k - 2$ cluster-level degrees of freedom for Pearson correlations and is approximate.

centering_rows

A string specifying which rows are used to estimate cluster means for within- and between-cluster decomposition. "pairwise_complete" uses only rows where both variables in the current pair are observed. "all_available" estimates each variable's cluster mean from all available rows for that variable, then correlates the pair on complete rows.

Details

Calculates bivariate within- and between-cluster correlations for clustered data, such as repeated measures nested in persons, dyads, teams, or other groups. Only recommended for continuous or binary variables.

For every variable pair, correlations are computed on rows where both variables and the cluster variable are observed. By default, `centering_rows = "pairwise_complete"` also estimates cluster means from this same complete-pair row set. This keeps the within residuals centered for the actual pairwise sample and makes the between correlation a correlation of matched pair-specific cluster means.

With `centering_rows = "all_available"`, each variable's cluster mean is estimated from all available rows for that variable before the pairwise correlation is computed. This can make the cluster means more stable when data are missing. It also mirrors a common multilevel-model preprocessing workflow, where person means are often created before the model applies complete-case filtering. That workflow is defensible in multilevel models. In `wbCorr`, however, the variables are treated symmetrically as a descriptive bivariate decomposition, so all-available centering means the two cluster means in a pair may be based on different occasions. For that reason, "pairwise_complete" is the default.

The within-cluster correlation is the pooled residual correlation. For a given pair, each observed value is centered around its cluster mean for that same complete-pair row set, and the correlation is computed on the resulting residuals. For Pearson within-cluster correlations, analytic inference uses $N_{\text{pair}} - k_{\text{pair}} - 1$ degrees of freedom, where N_{pair} is the number of complete observation pairs and k_{pair} is the number of clusters contributing at least one complete pair.

The between-cluster correlation is computed from pair-specific cluster means. With `between_weighting = "equal_clusters"`, every cluster contributes one equally weighted mean. With `between_weighting = "cluster_size"`, cluster means are weighted by the number of complete observation pairs in that cluster. Analytic p-values and confidence intervals for cluster-size weighted between correlations are approximate; use `between_inference = "none"` to report only the weighted coefficient.

Inspired by the `psych::statsBy` function, `wbCorr` allows you to calculate, extract, and plot within- and between-cluster correlations for further analysis.

Value

A `wbCorr` object that contains within- and between-cluster statistics. Use the `get_table()` function on the `wbCorr` object to retrieve a list of the full correlation tables. Use the `summary()` or `get_matrix()`

function on the wbCorr object to retrieve various correlation matrices, including ICCs in the merged ones. Use `get_ICC()` in order to get all intra class correlations (ICC(1,1)). Finally, use `to_excel()` on a table or matrix (or list of matrices) to save them.

See Also

[get_table](#), [summary](#), [get_ICC](#), [plot](#), [to_excel](#)

Examples

```
# importing our simulated example dataset with pre-specified within- and between- correlations
data("simdat_intensive_longitudinal")

# create a wbCorr object:
correlations <- wbCorr(simdat_intensive_longitudinal,
  'participantID')

# optionally compute sample-size weighted between-cluster correlations:
weighted_correlations <- wbCorr(simdat_intensive_longitudinal,
  'participantID',
  between_weighting = 'cluster_size')

# optionally estimate cluster means from all rows available for each variable:
all_available_correlations <- wbCorr(simdat_intensive_longitudinal,
  'participantID',
  centering_rows = 'all_available')

# returns a list with full detailed tables of the correlations:
tables <- get_table(correlations) # the get_tables() function is equivalent
print(tables)

# returns a correlation matrix with stars for p-values:
matrices <- summary(correlations) # the get_matrix() and get_matrices() functions are equivalent
print(matrices)

# Plot the centered variables against each other
plot(correlations, 'within')
plot(correlations, which = 'b')

# Store the list of correlation matrices to excel
to_excel(matrices)
```

wbCorr-class

wbCorr Class

Description

A class representing within- and between-cluster correlations.

Details

The wbCorr class is used to store within- and between-cluster correlations and provides methods for printing and summarizing the correlations.

See Also

[wbCorr](#)

Index

`get_ICC`, [2](#), [12](#)
`get_icc` (`get_ICC`), [2](#)
`get_ICCs` (`get_ICC`), [2](#)
`get_matrices` (`get_matrix`), [3](#)
`get_matrix`, [3](#), [8](#)
`get_table`, [4](#), [12](#)
`get_tables`, [3](#), [8](#)
`get_tables` (`get_table`), [4](#)

`plot`, [12](#)
`plot`, `wbCorr`-method, [5](#)
`plot.wbCorr` (`plot`, `wbCorr`-method), [5](#)
`print`, `wbCorr`-method, [6](#)
`print.wbCorr`, [7](#)
`print.wbCorr` (`print`, `wbCorr`-method), [6](#)

`show`, `wbCorr`-method, [7](#)
`show.wbCorr` (`show`, `wbCorr`-method), [7](#)
`simdat_intensive_longitudinal`, [7](#)
`summary`, [4](#), [12](#)
`summary`, `wbCorr`-method (`get_matrix`), [3](#)
`summary.wbCorr` (`get_matrix`), [3](#)

`to_excel`, [8](#), [12](#)

`update_wbCorr`, [9](#)

`wbCorr`, [2–4](#), [6–8](#), [9](#), [13](#)
`wbcorr` (`wbCorr`), [9](#)
`wbCorr`-class, [12](#)